SHABNA SHAIK

Software Developer

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aws certified
Developer
Associate

PROFESSIONAL SUMMARY

- Extensive experience in designing and developing **enterprise-grade distributed systems** and **scalable web applications** using **Java, Spring Boot, Node.js, React, AWS, and Azure**.
- Strong expertise in **microservices** and **event-driven architectures** using **Apache Kafka** for **high-volume, real-time data processing** systems.
- Experienced in designing and developing **RESTful APIs** and **GraphQL APIs**, including **Backend-for-Frontend (BFF)** patterns.
- Skilled in building **responsive front-end applications** using **React and TypeScript**, and backend systems using **Java and Node.js**.
- Hands-on experience with **relational and NoSQL databases** including **PostgreSQL, Azure Cosmos DB, and OpenSearch**.
- Proficient in **cloud deployments** on **AWS** and **Kubernetes (WCNP)** using **Docker**.
- Strong expertise in **real-time data processing** and **streaming architectures** using **Apache Kafka and Apache Spark**.
- Experienced in **CI/CD pipelines** using **Jenkins, KITT, GitLab CI/CD**, and enterprise DevOps tools.
- Configured **Prometheus, Grafana**, and **OpenObserve** for **monitoring, visualization, and log aggregation**.
- Experience in **API performance optimization** using **caching, indexing, and query tuning** techniques.
- Strong focus on **code quality** using **SonarQube, unit testing, and integration testing**.
- Experienced in using **AI-assisted development tools** such as **GitHub Copilot and Cursor** to enhance productivity and code quality.
- Experienced in **Agile/Scrum methodology** with **continuous delivery** practices.
- Strong experience in building **scalable, fault-tolerant systems** with emphasis on **performance, observability, and reliability**.
- Strong experience in **system design and scalability**, including **high availability, fault tolerance, and distributed system design principles**.

TECHNICAL SKILLS

CATEGORY	SKILLS
Programming Languages	Java, TypeScript, JavaScript, SQL, GraphQL
Frameworks	Spring Boot, Hibernate, React.js, Node.js, REST APIs, Microservices Architecture
Databases & Messaging	MySQL, PostgreSQL, Oracle, SQL Server, Azure Cosmos DB, Schema Design, OpenSearch, Apache Kafka, Apache Zookeeper
Cloud & DevOps	AWS (EC2, S3, RDS, Lambda), Microsoft Azure, Kubernetes (WCNP), Docker, CI/CD Pipelines, Looper, KITT, Concord, Helm, GitOps, Service Registry
Testing & Quality	JUnit 5, TestNG, Mockito, JaCoCo, SonarQube, Integration Testing, Data-Driven Testing
Software Development Practices	Test-Driven Development (TDD), Software Development Life Cycle (SDLC)
Monitoring & Observability	Prometheus, Grafana, Dynatrace, OpenTelemetry, OpenObserve
Tools & Methodologies	Maven, Git, Agile/Scrum, Event-Driven Architecture, Domain-Driven Design, Circuit Breaker Pattern, GitHub Copilot, Cursor

PROFESSIONAL EXPERIENCE

Software Developer

Walmart Global Tech
Apr 2025 – Present
Sunnyvale, CA.

Project Description:
Worked on **Seller Order Lifecycle systems** using **microservices and event-driven architecture**, supporting **order processing, disputes, returns, shipment tracking, and carrier update processing** across the **US market**.

Key Roles and Responsibilities:

Front-End Development:

- Collaborated with frontend teams to design and document **GraphQL schemas and REST API contracts**

- Developed API response structures optimized for frontend consumption using **Backend-for-Frontend (BFF)** pattern
- Designed RESTful APIs for **order lifecycle and shipment tracking integrations.**
- Implemented **pagination and filtering** for efficient and scalable data retrieval.
- Built **real-time tracking status APIs** consumed by customer-facing applications
- Enabled **correlation ID propagation** for end-to-end request tracing across services

Back-End Development:

- Designed and developed **RESTful and GraphQL APIs** using Spring Boot 3.x and Java 21 for **dispute management, order processing, and package tracking services**
- Implemented **event-driven architecture using Apache Kafka** for **order lifecycle and package tracking events**, enabling scalable and reliable asynchronous processing
- Designed and optimized **Azure Cosmos DB data models** for high-volume read/write workloads
- Developed **OpenSearch integration** enabling full-text search for package tracking with **low-latency query responses**
- Implemented **Domain-Driven Design (DDD)** with clear separation of concerns using persona, adapter, and common modules
- Developed **batch processing jobs** for scheduled reconciliation, cleanup, and data consistency operations

Cloud & DevOps:

- Containerized microservices using **Docker with multi-stage builds** and deployed on **Kubernetes (WCNP)**
- Built **CI/CD pipelines using KITT, Looper, and Concord** for automated deployments
- Implemented **GitOps practices with Service Registry (SR2)** and centralized configuration management (CCM2)
- Configured **monitoring and observability using Dynatrace, Prometheus, Grafana and Micrometer**
- Managed **Kafka topic configuration and partitioning** for scalable event processing
- Configured **multi-environment deployments (dev, stage, prod)** with approval workflows

Environment: Java, Spring Boot, GraphQL, REST APIs, Apache Kafka, Azure Cosmos DB, OpenSearch, Docker, Kubernetes, Dynatrace, Prometheus, JaCoCo, JUnit 5, TestNG, Mockito, Git and Grafana

Software Developer

Fresenius Medical Care
Feb 2024 – Mar 2025
Lawrence, MA.

Project Description:
Building a **next-generation patient management system** to optimize healthcare workflows, featuring secure data handling and seamless integration with medical devices. The solution ensures compliance with **HIPAA standards** and provides real-time analytics for healthcare providers.

Key Roles and Responsibilities:

Front-End Development:

- Developed responsive UIs using **React**, **TypeScript**, and **Material UI**, enhancing user experience for healthcare providers.
- Implemented dynamic data visualizations using **D3.js**, improving patient data tracking across dashboards.
- Optimized performance by using **React's lazy loading** and **code splitting** for faster page loads.
- Created reusable **UI components**, reducing code redundancy and maintaining consistency across the platform.
- Integrated **React** with APIs to provide real-time data updates for healthcare management.

Back-End Development:

- Built **RESTful APIs** using **Spring Boot** and **Node.js** to manage patient data and integrate seamlessly with the front-end.
- Integrated **Salesforce** and **FHIR APIs** to enable interoperability with third-party healthcare systems.
- Implemented **encryption** and **role-based access control** to ensure secure handling of sensitive patient data.
- Optimized **PostgreSQL** database queries, reducing latency and improving data retrieval speed.
- Designed **microservices** to handle patient information processing efficiently and securely.

Cloud & DevOps:

- Deployed and monitored applications on **AWS EC2**, ensuring high availability and scalability of healthcare systems.
- Deployed **microservices** on **AWS EC2**, ensuring high availability and auto-scaling during peak usage.
- Automated **CI/CD pipelines** using **Jenkins** and **Docker**, streamlining deployments and testing.
- Used **AWS Lambda** for **serverless computing**, reducing infrastructure costs and operational overhead.
- Monitored application health with **AWS CloudWatch** and **ELK Stack** for proactive issue detection.
- Managed infrastructure as code (**IaC**) using **AWS CloudFormation** for efficient resource provisioning.
- Built Spark streaming jobs for real-time patient data analytics on **AWS EMR**.

- Developed ETL workflows with **Apache Spark** to preprocess and analyze patient records efficiently.

Environment: Java, Node.js, React, Spring Boot, Angular, AWS (EC2, Lambda, S3), PostgreSQL, Git, Jenkins and Apache Spark

Software Developer

Mastercard Global Payment Technology Solutions
Nov 2022 – Jan 2024
St Louis, MO.

Project Description:
Developed a **fraud detection and prevention platform** leveraging **machine learning models** to analyze transaction patterns and detect anomalies in real-time.

Key Roles and Responsibilities:

Front-End Development:

- Developed responsive dashboards with **React**, **Chart.js**, and **Material UI** for real-time fraud analytics.
- Integrated **WebSockets** to display live notifications for fraud alerts, improving user response time.
- Implemented **React hooks** and **functional components** to streamline state management and UI rendering.
- Enhanced UX/UI with custom **React** components for visualizing transaction anomalies.
- Optimized rendering performance with **React.memo** and **React.lazy** for large datasets.

Back-End Development:

- Built **Java-based microservices** using **Spring Boot** to process payment transactions and detect fraud patterns.
- Integrated **Apache Kafka** for real-time data streaming and transaction analysis.
- Utilized **Node.js** for building **API endpoints** to interact with front-end dashboards and fraud detection models.
- Secured APIs using **OAuth2** for safe access to payment data and fraud alerts.
- Implemented **Apache Spark-based fraud analytics pipeline** for large-scale transaction datasets.
- **Leveraged Spark SQL** to enhance **fraud detection query performance**, reducing response time.
- Implemented real-time data processing with **Apache Kafka** to reduce fraud detection latency.

Cloud & DevOps:

- Deployed microservices on **AWS ECS**, leveraging **Docker** for containerization and scalability.
- Automated deployments with **GitLab CI/CD** pipelines, reducing manual intervention and ensuring consistency.
- Utilized **AWS DynamoDB** to store transaction logs and enhance query performance for fraud detection.
- Configured **AWS CloudWatch** to monitor performance and set up alarms for transaction anomalies.
- **Deployed Spark jobs** on AWS EMR for batch processing of fraud detection data.
- Set up **auto-scaling groups** in **AWS** to handle traffic spikes during high transaction volumes.

Environment: Java, Node.js, React, Spring Boot, Kafka, Apache Spark, AWS (ECS, DynamoDB, S3), GitLab CI/CD

Software Developer

CarMax
Jan 2021 – Oct 2022
Atlanta, GA.

Project Description:
Developed a **vehicle inventory and customer service platform**, enabling seamless interaction between customers and dealership staff. This application included features for vehicle search, reservation, and financing options.

Key Roles and Responsibilities:

Front-End Development:

- Designed a dynamic and responsive **React-based UI** for vehicle search and customer interaction.
- Integrated **Google Maps API** with **React** to provide location-based vehicle inventory searches.
- Implemented **pagination** and **virtual scrolling** in **React** to efficiently manage large vehicle datasets.
- Developed **React components** for seamless user interaction in vehicle reservation and financing workflows.
- Enhanced UI accessibility using **ARIA** standards to ensure inclusivity and compliance with WCAG guidelines.

Back-End Development:

- Developed **Spring Boot-based REST APIs** for managing vehicle inventory, reservations, and payments.
- Implemented **Node.js** to handle asynchronous operations and integrate with third-party services (**payment gateways**).

- Built and optimized **PostgreSQL** queries to improve database access times and handle high traffic volumes.
- **Utilized Apache Spark** for vehicle data processing and customer insights analysis.
- **Implemented Spark SQL** for optimizing vehicle search queries, improving response time by 40%.
- Integrated **payment services** with secure **token-based authentication** and **PCI DSS** compliance.
- Developed background jobs for batch processing of inventory updates and customer communications.

Cloud & DevOps:

- Deployed services to **AWS Elastic Beanstalk**, simplifying the application lifecycle and scaling.
- Set up automated **backups** and **disaster recovery** solutions with **AWS RDS snapshots** and **S3**.
- Used **AWS CloudFront** for efficient content delivery and optimized page load times.
- Automated deployment processes using **Jenkins** for **CI/CD**, improving team productivity and reducing errors.
- Leveraged **AWS CloudWatch** for proactive monitoring and logging, ensuring uptime and reliability.

Environment: Java, Node.js, React, Spring Boot, Apache Spark, AWS (Elastic Beanstalk, CloudFront, RDS), PostgreSQL, Jenkins

Software Engineer

Grabhouse – Mumbai, IN
Mar 2017 – Dec 2020

Project Description:
Developed a **property rental platform** providing a seamless interface for landlords and tenants to list, search, and book rental properties without brokers.

Key Roles and Responsibilities:

Front-End Development:

- Designed interactive UIs with **ReactJS** for property listing, search, and booking functionalities.
- Integrated **third-party APIs** for real-time property listing updates and location-based search.
- Built reusable **React components** for consistency across the platform and simplified UI updates.
- Implemented **user authentication** and session management using **React Context** and **hooks**.
- Optimized **SEO** by rendering metadata dynamically, improving search engine visibility.

Back-End Development:

- Developed **Java-based Spring MVC services** for handling property listings, bookings, and payments.
- Built **Node.js APIs** to manage user interactions, including property search and reservation functions.
- **Implemented Apache Spark** for analyzing rental trends and optimizing pricing algorithms.
- Processed large-scale rental transaction data using **Spark on AWS EMR** for predictive analytics.
- Integrated **cron jobs** for scheduled property data synchronization and notification dispatch.
- Enhanced data integrity and security with **JWT-based authentication** and secure data storage.
- Built an **event-driven architecture** to handle high-volume property searches and real-time updates.

Cloud & DevOps:

- Hosted services on **AWS EC2** with load balancers to ensure application scalability and availability.
- Used **S3** for storing property images and user-generated content, ensuring durability and performance.
- Managed **infrastructure provisioning** and **deployment automation** with **Terraform** and **Jenkins**.
- Monitored application health and performance using **AWS CloudWatch**, ensuring uptime and reliability.
- Implemented **AWS RDS** for managed database services and automated backups.

Environment: Java, Node.js, React, Spring MVC, AWS (EC2, S3, RDS), MySQL, Jenkins and Apache Spark

EDUCATION

- **Master's in Computer Science –** Northwest Missouri State University, Maryville, Missouri**.**
- **Bachelor of Technology in Computer Science** – Sree Vidyanikethan Engineering College, Tirupati, India.

CERTIFICATIONS

- **AWS Certified Developer – Associate**